

Kyle Semple

Forward Deployed Engineer | Full-Stack Geospatial and Data Systems

Technical Systems Translator | AI Workflows | Operational Reliability

Ann Arbor, MI | 734-882-9095 | kmsemp26@gmail.com | Portfolio | GitHub | LinkedIn

PROFILE

Implementation-oriented engineer who translates ambiguous client and operational requirements into working software, validated data workflows, and supportable handoffs. Builds across frontend, backend, spatial data, QA, and deployment controls, with experience carrying messy physical-world data from source review through usable systems.

CORE: Forward-deployed delivery | Full-stack implementation | Geospatial data engineering | Reliability and validation | Client translation | Technical documentation

TECHNICAL SKILLS

Applications: React, TypeScript, Python, FastAPI, Pydantic, REST APIs, Node/Express, Vite

Data and geospatial: PostgreSQL, PostGIS, SQL, ArcGIS, GDAL/ogr2ogr, GeoJSON, GeoTIFF, GeoParquet, Leaflet, IndexedDB

Delivery and reliability: Docker, Cloud Run, Supabase, GitHub Actions, browser/backend testing, manifests, SHA-256 provenance, fault handling

SELECTED SYSTEMS

HPS Geospatial Platform

[React/TypeScript](#) | [Python/FastAPI](#) | [PostgreSQL/PostGIS](#) | [Geospatial production and validation](#) | [Live](#) | [Case Study](#)

- Stabilized the core platform by resolving 17 pre-existing browser regressions; the reliability phase passed 98 browser tests, 228 backend tests, TypeScript, ESLint, and the production build.
- Delivered a five-minute, single-use Studio-to-Georeferencer handoff with explicit failure states, manual fallback, worker isolation, timeouts, and manifest-backed provenance. The local synthetic PNG-to-Recovery-to-GeoTIFF flow passed a 33/33 browser matrix and 266 backend tests; live production-database behavior was outside the retained evidence.
- Generalized country-aware river artifacts beyond Guyana and audited 26 country entries - 5 verified, 10 partial, 10 unavailable, and 1 retained legacy result - with 22 packaged artifacts, 4 documented withheld entries, hashes, and registry/manifest/runtime checks.

Portfolio 2.0 and Digital Twin AI Agent

[React/TypeScript](#) | [Gemini proxy](#) | [CI/governance](#) | [Runtime failure handling](#) | [Live](#) | [Build Case Study](#) | [Agent Case Study](#) | [GitHub](#)

- Built and maintained a public engineering evidence system across 400+ commits with CI gates, route validation, secret scanning, ADRs, threat modeling, AI attribution records, and implementation documentation.
- Implemented a scoped assistant behind a server-side Gemini proxy with prompt-injection filtering, approved command routing, request and input limits, history controls, and human handoff; later added distinct handling for rejected origins, missing configuration, oversized input, and backend outages plus deployment-safe Docker and Cloud Build configuration.

Guynode Spatial Data Hub

[Spatial data migration](#) | [Static web artifacts](#) | [Dataset governance](#) | [Citation and provenance](#) | [Live](#) | [Case Study](#)

- Delivered the client-approved replacement for a legacy spatial hub by converting 34 ESRI Shapefiles into web-optimized GeoJSON and reorganizing 85 datasets across 7 categories with validation, checksums, hosted-asset governance, map previews, and citation/provenance tooling.

PROFESSIONAL EXPERIENCE

HPS Geospatial Consulting LLC - Independent Systems Consultant

Oct 2021-Present

[Client-facing GIS, AI workflow, and technical implementation](#)

- Translate unclear requirements, data limitations, and user needs into spatial workflows, implementation documentation, and handoff-ready systems; supported MOH GIS delivery through ArcGIS Online, Survey123, Dashboards, Experience Builder, troubleshooting, and stakeholder guidance.

Tech Providers, Inc. - GIS Analyst, Consumers Energy

Mar-Dec 2024

- Reconciled gas-distribution records across ArcMap, SAP, work-order systems, MicroStation drawings, and utility source materials; applied repeatable QA checks and escalated ambiguous cases before production GIS corrections.

Apex Systems - Quality Control Specialist, CenterPoint Energy

Sep 2022-Dec 2023

- Maintained production-quality accuracy across 120+ weekly electric-utility service requests through source-record review, repeatable QA checks, discrepancy resolution, and escalation before final ArcMap edits.

EDUCATION AND CREDENTIALS

B.A., Geography - Queen's University | Kingston, Ontario | Sep 2016-Apr 2021

Selected credentials: Google Project Management Professional Certificate | Google Data Analytics